

ACEnet Research Fellowships

Application Guide and Form

Round II

This document contains the application guide and form for applying for research personnel support from ACEnet in Round II. Please obtain the most recent version of the guide and form from www.ace-net.ca.

1. Overview of the ACEnet Research Fellowships (ARF) Program

The program has two objectives:

- Increase the number and quality of research personnel involved in high-performance computing (HPC) at ACEnet Member Institutions.
- Facilitate the optimal exploitation of ACEnet's computing infrastructure, leading to research discoveries of international significance.

A total of up to \$1M will be awarded in Round II. Financial support is available for three types of research fellowships:

- **Senior Visiting Fellowship:** To attract and support world-class researchers who wish to make short term visits to an ACEnet institution. The Fellowships may also be used to support visiting faculty on sabbatical leave from their home institution.
- **Postdoctoral Fellowship:** To assist in attracting the best postdoctoral fellows to work with ACEnet researchers. In order to be eligible for a postdoctoral fellowship the candidate must have completed their PhD within the last 5 years prior to the commencement of the Fellowship.
- **Graduate Fellowship:** To assist in attracting the best graduate students to work with ACEnet researchers.

	Min/Max Term	Maximum ACEnet Eligible Costs	ACEnet Contribution (%)	Matching Contribution (%)	Allowable Expenses
Senior Visiting Fellowship	1 to 8 weeks	\$10,000 (\$2K for travel plus \$1K/week)	\$5,000 (50%)	\$5,000 (50%)	travel, lodging, subsistence
Postdoctoral Fellowship	1 or 2 years	\$40,000/yr	\$20,000/y (50%)	\$20,000/y (50%)	salary, benefits only
Graduate Fellowship	1 or 2 years	\$20,000/yr	\$10,000/y (50%)	\$10,000/y (50%)	salary, benefits only

Table 1: Financial Details of the Fellowships.

The applications are due March 29, 2009, the results will be announced June 19, 2009 and the candidates must take up the fellowship by January 15, 2010.

2. How to Apply

Submit the application form and requested material electronically. The web form can be found at <http://fellowships.ace-net.ca>. Instructions for completing the form are located at the top of the page. All applications will be treated as confidential. Information on the application process is given below. Questions regarding the application process may be directed by email to the ACEnet main office (fellowships@ace-net.ca).

Eligibility

Only individual ACEnet Project Leaders actively using ACEnet's facilities, who are full-time tenured or tenure-track faculty at ACEnet Member Institutions as of March 1, 2009 are eligible to apply. The ACEnet Member Institutions are Acadia University, Cape Breton University, Dalhousie University, Memorial University of Newfoundland, Mount Allison, Mount St Vincent University, St Francis Xavier University, St Mary's University, University of New Brunswick and University of Prince Edward Island.

Identity of the Candidate

For Senior Visiting Fellowships, the applicant must nominate a specific visitor at the time of application.

For Postdoctoral and Graduate Student Fellowships, the identity of the candidate need not necessarily be known at the time of application. If an application is successful and either the candidate is not known at the time of application, or the candidate listed in the application withdraws, then the applicant will be given up to November 2, 2009 to identify a suitable candidate and submit the candidate's curriculum vitae for decision by the ACEnet Fellowship Committee. The quality of the candidate is an important criterion in determining the success of an application. Successful candidates must begin their Fellowship by January 15, 2010.

Curriculum Vitae

When providing CV's, no specific format is required. We suggest submitting an NSERC Form 100, Parts I and II for the curriculum vitae of applicants and candidates (senior visitors and postdoctoral fellows). However, any reasonable document with equivalent information is acceptable. Information on quality of graduate student applicants should be provided, if possible.

Project Description

The overarching criterion determining the success of an application is its potential to generate high-impact research results that will advance ACEnet's status as a world-class centre for high-performance computing.

Please address the following points when preparing your project description:

- Applicants who received funding in Round I must describe how the supported visitor, postdoctoral fellow or graduate student made use of ACEnet resources. A summary of the main results and achievements of the Round I project is also required.
- Explain how the proposed research and training will benefit directly from access to ACEnet's HPC resources.
- Projects that bring expertise and techniques into ACEnet that are likely to benefit other researchers are especially valuable. Where possible, describe the experience of the candidate in HPC.
- Applicants should specify the anticipated outcomes in concrete terms. For example, give the names of journals in which you anticipate publishing results, or quantify the degree to which the proposed work is world-class.
- If appropriate, describe the potential for development of intellectual property, or new relationships with private sector firms.
- Applicants should estimate the computational needs (e.g., number of processor-hours) of the proposed research. Ambitious projects that will establish new standards within their field are encouraged. On the other hand, if the success of a project requires all of ACEnet's processing power for a year, it is not likely to be funded. Remember that ACEnet is a shared resource.

Note the page length restrictions on the application form. Any submitted material in excess of the stated maxima will be ignored.

Multiple Applications

Project leaders may normally submit no more than two applications. However, applicants should be aware that in the case of multiple applications the success of one application will be taken into consideration by the ACEnet Fellowship Committee when considering the remaining applications. Requests for extensions of Round I Fellowships will be treated as separate applications (see below).

Extensions of Fellows Funded in Round 1

The ACEnet Fellowship Committee will give priority to new awardees and PhD students just completing the second year of a Round I ACEnet Fellowship. The Committee will consider awarding one year extensions of Round I ACEnet postdoctoral fellows if a strong case can be made for the quality of their scientific research and the effective use of ACEnet research facilities.

Proposed Budgets

Values of the Fellowship in Table I are upper bounds. You should apply for the actual amount needed to support your project up to the maximum allowed.

Matching Funds and Eligible Expenses

All Fellowships typically require the applicant to contribute matching funds: for every \$1 of matching funds, \$1 of ACEnet funds is available, up to the indicated maxima (Table 1). The key criterion for deciding if matching funds are eligible is that they fund people to do academic research. Examples of eligible matching funds are:

- NSERC funds paid to a student or postdoctoral fellow.
- Internal or external scholarships paid to graduate students.
- Funds provided by a private sector organization, partnered with an ACEnet project Leader.
- Faculty start-up grants.

Ineligible sources include:

- Payments for contract research where the resulting intellectual property is owned by a private firm.
- Teaching assistance support of graduate students.

Note that matching funds must be spent on the eligible expenses shown in Table I. For example, NSERC equipment grant funds used to provide a workstation for use by a postdoctoral fellow are not eligible for use as matching funds for an ACEnet Postdoctoral Fellowship.

Relationship Between NSERC, CIHR or SSHRC Awards and ACEnet Fellowships

ACEnet Fellowship Committee encourages applications from graduate students and postdoctoral fellows about to take up a new major award from NSERC, CIHR or SSHRC (or similar funding agency), provided the proposed research and the candidate's qualifications satisfy the selection criteria. Such awards will be given a high priority by the ACEnet Fellowship Committee but its decision will be contingent on the applicant obtaining the external award. The ACEnet funding will not be awarded to top up existing external awards or transferred to another candidate. The ACEnet contribution may reach a maximum of \$10,000/y in the case of graduate students and \$20,000/y for postdoctoral fellows (Table 1).

ACEnet must be informed of any existing or pending awards at the time of application or when the candidate is approved. In such cases the applicant will be asked to justify the amount requested from ACEnet.

Duration of Awards

Senior Visiting Fellowships can be funded for one to eight weeks; postdoctoral fellows and graduate students are eligible for funding for one to two years (see Table 1). Masters students will not be eligible for funding beyond the second year of their graduate program and PhD students beyond the fourth year of their program.

The start date for commencement of ACEnet funding may also be set after the announcement of the award (retroactively) but not prior to April 1, 2009. Any eligible funds expended on the Fellowship candidate after the project commencement date are eligible for matching.

3. Outline of Review Process

The ACEnet Fellowship Committee (AFC) makes recommendations for awards with input from the Fellowship Review Committee (FRC). The AFC's recommendations are submitted to the Board of Directors through the Research Directorate for approval.

The FRC is an external review panel chaired by the Chair of the ACEnet Fellowship Committee or designate. The external membership of the FRC is composed of experts whose research interests reflect the range of proposals submitted in Round II. The membership of the FRC is determined by the Board of Directors on the recommendation of the AFC. The role of the FRC is to determine the excellence of the proposed research in the applications within the context of the objectives stated in Section 1.

Based on the merit of the proposed research within the context of the program goals, the FRC determines which of the applications submitted are acceptable and merit funding and ranks each of the acceptable applications within each category of Fellowship. This ranking, along with appropriate comments, is submitted to the AFC.

The AFC reviews the ranking of the FRC and makes a recommendation to the Board of Directors on which of the applications deemed acceptable by the FRC to fund. This recommendation is based on the following criteria

1. Excellence of the candidate (if known) and the proposed research (as determined by the FRC).
2. Potential to bring new expertise, methodologies and/or applications in high-performance computing into the ACEnet community.¹
3. Appropriateness of and justification for the proposed budget.
4. The AFC will also take into account the distribution of Fellowships across the Member Institutes and research disciplines.
5. The research accomplishments of candidates funded under Round I using ACEnet's computational resources.

Applicants should estimate the computational needs (e.g., number of processor-hours) of the proposed research. Ambitious projects that will establish new standards within their field are encouraged. On the other hand, if the success of a project requires all of ACEnet's processing power for a year, it is not likely to be funded. Remember that ACEnet is a shared resource.

¹ The AFC will not normally award a Postdoctoral Fellowship to a candidate who has been employed by the applicant as a postdoctoral research assistant or fellow for more than three years prior to the application, unless the applicant can clearly demonstrate how the proposed project and the candidate satisfy this criterion.

In preparing its recommendation the AFC will take into account the budget made available from the ACEnet Sustaining Funds by the Board of Directors. In all cases, only those applications deemed acceptable by the FRC will be recommended to the Board of Directors.

4. Summary of Timelines and Deadlines

February 2009	Call for Proposals.
March 29, 2009	Applications Due. No exceptions or extensions.
June 19, 2009	Announcement of Results.
November 2, 2009	Identity of PDF or grad student finalized.
January 15, 2010	Deadline for candidate to start.

Table 2: Timeline and Deadlines.

We anticipate that, depending on availability of funds, there will be a second competition in Round II, with a call for proposals in January 2010.

5. Announcement of Awards and Transfer of Funds

As soon as the Review Process is completed, the Project Leader (PL) will be notified by e-mail that they have been successful. Following the e-mail, a letter outlining the terms and conditions of the award will be sent to the PL and an award agreement will be sent to the Research Office at the PL's institution. Applicants must belong to an ACEnet Member Institute (MI) and no funds will be sent to any organization other than ACEnet MI's and no funds will be transferred without a signed and completed "Notice of Award" agreement.

The terms and conditions governing the award are:

- The Candidate must be identified and approved by the ACEnet Fellowship Committee before any funds are transferred. If the postdoctoral fellow or graduate student was not known at the time of the application, the PL must provide information on the candidate (include a curriculum vitae for postdoctoral fellows) by November 2, 2009. This information will be reviewed by the AFC and funding may be withheld if the candidate is considered unacceptable. The awardee will be asked to confirm if any scholarship or fellowship support is to be received during their tenure as an ACEnet Fellow. If so, the value of ACEnet support may be revised.
- Only the salary expenses are eligible for Postdoctoral Fellowships and Graduate Student awards. For Senior Visiting Fellowship awards, Travel, Lodging and Subsistence costs are eligible to the amounts outlined in Table 1.
- ACEnet's contribution to the project must be matched by at least 50% with funds from other sources (as specified).

- Any changes to the project with respect to candidate or project scope, objectives or methodology must be approved by ACEnet.
- The Notice of Award agreement must be signed and the project must start by January 15, 2010.
- A statement that ACEnet and Sun Microsystems will announce the awards under this program at a later date; therefore, this award should not be publicly announced without prior permission.
- Inter-university transfers are not permitted.
- All Project Leaders are required to submit a report describing the research accomplished prior to the release of year 2 funds (if any), or immediately following the support period for periods less than one year. The report must describe the use of ACEnet's HPC facilities. Projects are also expected to acknowledge ACEnet support in all journal publications, oral presentations, and posters, etc., describing their research.
- All ACEnet Fellows (Research and Postdoctoral) are required to participate in ACEnet events during the term of this award. These events include seminars, ACEnet Institute activities, where appropriate, and other ACEnet technical and research meetings. Information of participation in such events should appear in the reports on the Fellow's progress (see above).
- There is no carry-forward provision for unspent funds.
- ACEnet will require financial reports on the expenditures of the project on December 31, 2009, June 30, 2010 and December 31, 2010.
- While second year funding has been approved, transfer of the second year's funds will be contingent on all reporting obligations and participation obligations being met.
- Reviewers' comments will not be made available and there is no appeal process.